

ECE 482

4-Bit Multiplier in 45nm Technology

Ashton Billings, Albus Lai, Chirag Maheshwari
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1 Overview

In this project, we were tasked with laying out a 4-bit multiplier circuit in 45nm technology with the explicit goal of making it as low power as possible. In order to accomplish this, we carefully considered possible variants with each block, going with what we believed to be either the lowest power option, or easiest to work with in a low power system. Shown below is our design block diagram:

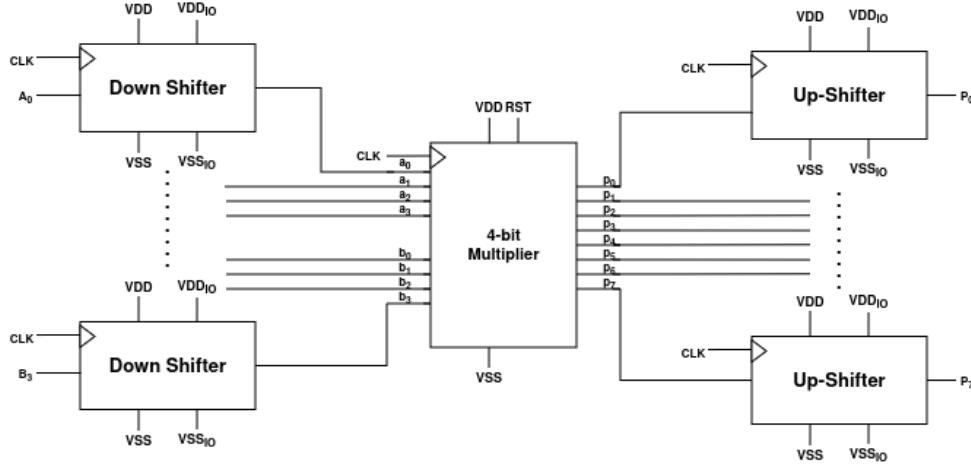


Figure 1: Top-level block diagram.

Our design uses level-shifters, superbuffers, registers, supporting circuitry such as inverters and buffers, and of course, the 4-bit multiplier. More detail on the specifics of each block is below. The main power reduction strategy we used here was designing our 4-bit multiplier in 1.1V transistors rather than 1.8V's, the same as our inputs and outputs; translating between the two via level-shifting. We also implemented pipelining as well allowing parts our circuits to operate even slower under the same time requirement. There were even further tweaks and considerations to reduce power on the block level, this will be discussed on each individual block below.

One final note about our report is there was a bit of uncertainty about the output ready requirement. Our initial understanding of the 3 clock cycle requirement is that upon the first clock cycle our inputs were ready, our output should be ready 3 cycles later. For each testbench then, this meant our outputs need to be ready by 40ns. Since the input and output need to be clocked, we assumed this left us with 1 register in our multiplier to pipeline. Upon discussion with the TAs, however, we were told 2 registers was fine for our multiplier for a total of 4 registers. This would mean our output values need to be ready by 44ns instead. Due to this confusion, to be conservative, we included results from two VDD operating voltages of 1.1V and 1.05V, in which the 1.1V outputs are ready by 40ns, while the 1.05V are ready by 44ns.

2 Gate Level Diagrams

2.1 4-Bit Multiplier

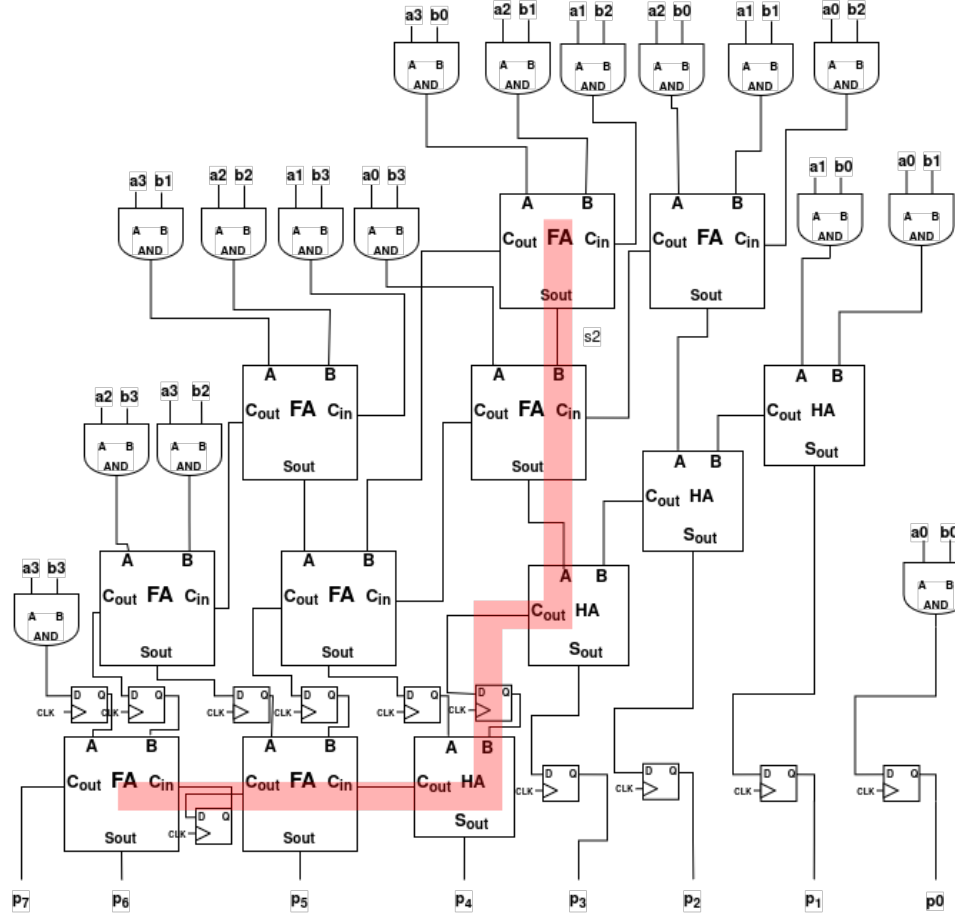


Figure 2: 4-Bit Multiplier Gate Level

The design we went with is different from the standard 4-bit multiplier. This design is inspired by a carry-save adder. With this design, the number of adders on the critical path is 6 (as marked by the red path), which is 2 fewer than that of a standard 4-bit multiplier design. Additionally, with this design, the input arrival times at each adder should be closer together, thereby reducing glitches. Next, a shorter critical path would theoretically allow a greater drop in voltage than a standard design. Lastly, the pipeline placement was to cut the critical path evenly, such that the critical path between each register stage is 3.

2.2 Half-Adder

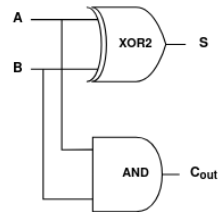


Figure 3: Half-Adder Gate Level

On the gate level, our half-adder is the typical design. Which logic family was implemented for each gate will be seen in the schematic section.

2.3 Full-Adder

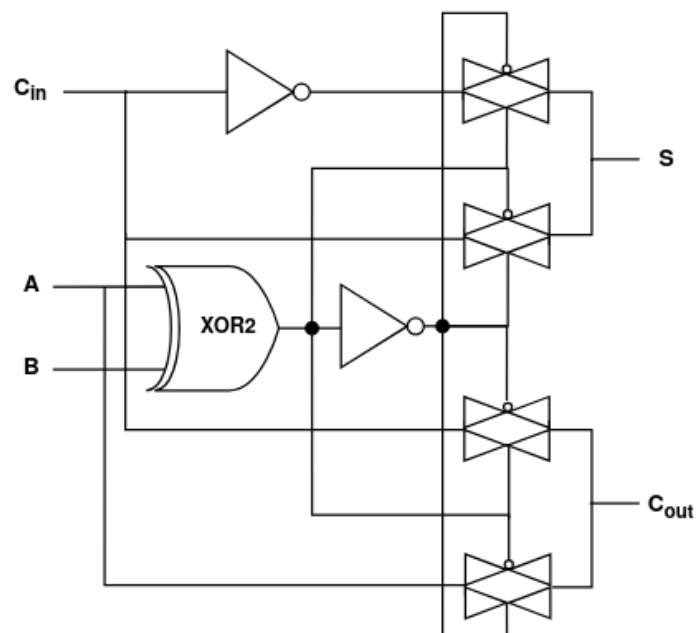


Figure 4: Full-Adder Gate Level

We use a modified transmission gate based full adder. We use an XOR2 to create our propagate signal $P = A \oplus B$. For the sum output, P is used as a select signal for a transmission gate multiplexer that selects between C_{in} and $\overline{C_{in}}$, implementing $S = P \oplus C_{in} = A \oplus B \oplus C_{in}$. For the carry output, we use transmission gate logic controlled by $\overline{A}$ and $\overline{C_{in}}$ to implement C_{out} . For simplicity, we moved the XOR2 inverter outside the XOR2 cell to avoid having to layout another cell while implementing correct logic.

2.4 Down-Shifter

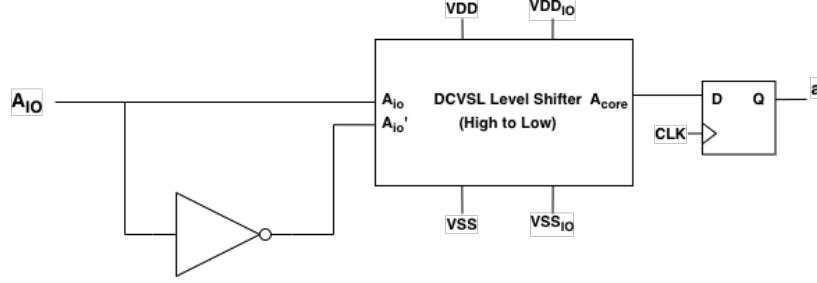


Figure 5: Down-Shifter Gate Level

2.5 Up-Shifter

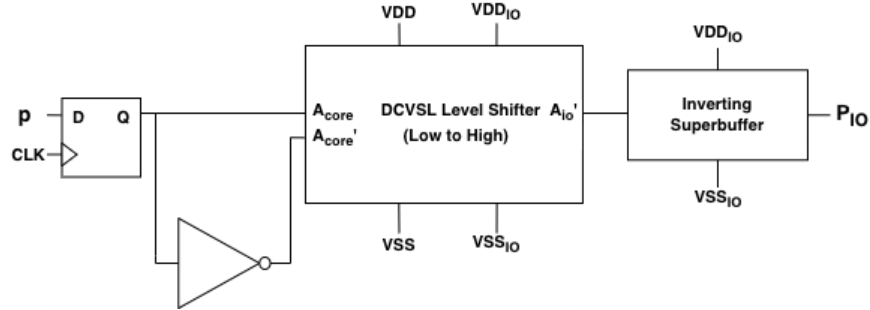


Figure 6: Up-Shifter Gate Level

3 Latency

3.1 Latency Figures

In both test benches, the inputs get clocked at around 28ns. In both cases, we see that the inputs are correct for each test vector, with the outputs within $\pm 10\%$ at 40ns. Given our 250MHz operating frequency giving us a 4ns period, we see that our outputs arrive in time exactly 3 clock cycles later.

In order to achieve this, our design used three pipeline stages, with one register at the end of our down-shifter, one register in the middle of our multiply unit, and another register at the beginning of our up-shifter. Starting the clock at 28ns after the first register, there are two more register stages our values must go through. By having two more stages rather than say three, we allow 4ns of time after the output of the last register to hit our desired value. This slack allowed us to make a smaller superbuffer than what an larger and less delay having superbuffer would be, greatly reducing area and therefore power.

4 Power Reduction Strategy

Below is an exhaustive list of power reduction strategies employed. More details on each can be seen throughout the report as well.

4.1 Methods Used

- (a) **Pipelining:** By pipelining our multiplier circuit, we allow the individual parts of our circuit to operate slower and therefore be smaller and consume less power.
- (b) **Level-Shifting:** By levelshifting down our voltage to 1.1V for multiplier operation, we greatly reduce our switching power. This reduction is so great due to the fact that switching power is related by the following expressions: $P_{dynamic} = \alpha fCV^2$. Due to the square relationship on voltage, and reduction in power will give benefits to the power of two.
- (c) **Sizing Considerations:** We gave careful considering during the schematic phase on the sizing of certain elements. We started with the naive solution of simply designing for minimum size, but soon realized that there may actually be power benefits in sizing up a bit. By making our cells bigger than minimum size, they could actually operate faster and therefore we could lower the operating voltage of 1.1V even lower.
- (d) **Logic Reconfiguring:** Rather than go with standard designs for each block, we considered alternatives to minimize logic blocks, delay, and power. Such alternatives are as follows: our multiplier was altered to minimize critical path delay. Then for each part of the multiplier, many alternatives were considered for the schematic side. Our register is a c^2mos design, to remove any skewed clock issues with short circuiting, which would increase dynamic power. For our up and down shifters, as seen in our circuit diagram, we used the notted output of our level-shifters in conjunction with our superbuffers. This was done purposefully as our level-shifters naturally have notted outputs available, and our superbuffers design optimally works with odd inverters. For our AND gates, we tested both transmission gate logic and cmos logic, finding cmos logic to consume less power.
- (e) **Drain Source Merging:** For many of our blocks, we designed them such that minimal diffusion regions were used, as will be seen in our layout section. Of particular note is the AND and register layouts, which only use one diffusion region. This minimizes parasitic capacitance's.

5 Transistor-Level Schematics

5.1 4-Bit Multiplier

5.1.1 XOR2

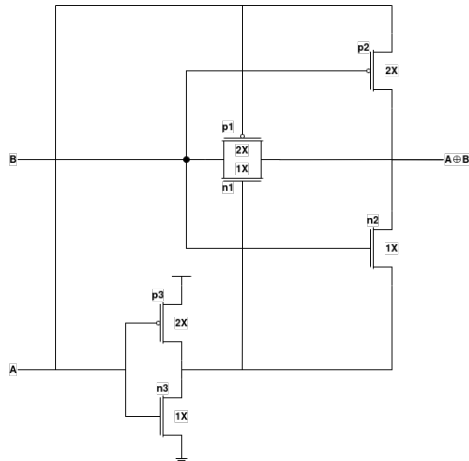


Figure 7: XOR2 Schematic

Device	Width (nm)	Length (nm)
pmoslv	240	45
nmoslv	120	45

Table 1: XOR2 Transistor Dimensions

5.1.2 AND2

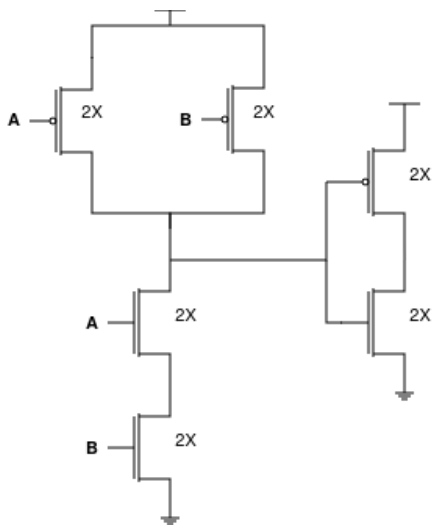


Figure 8: AND2 Schematic

Gate Type	Device	Width (nm)	Length (nm)
Inverter	pmos1v	240	45
	nmos1v	240	45
NAND2	pmos1v	240	45
	nmos1v	240	45

Table 2: AND2 Transistor Dimensions

5.1.3 Register

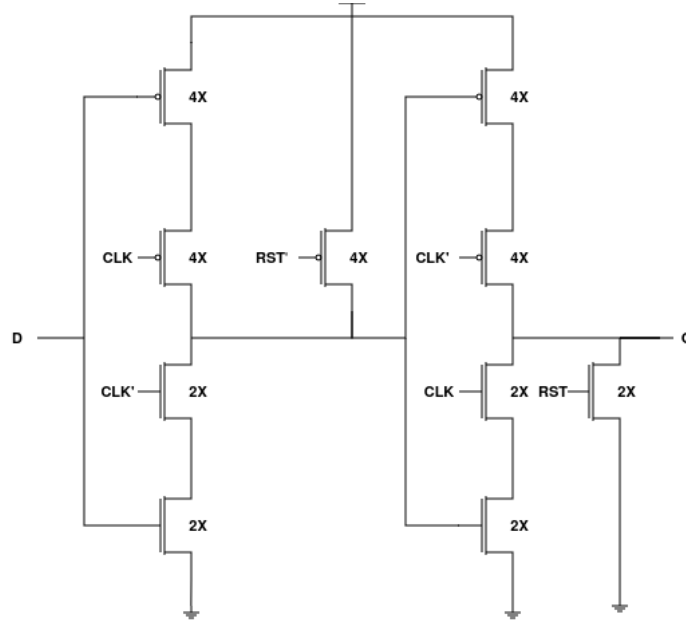


Figure 9: Register Schematic

Device	Width (nm)	Length (nm)
pmos1v	480	45
nmos1v	240	45

Table 3: Register Transistor Dimensions

5.1.4 Transmission Gate

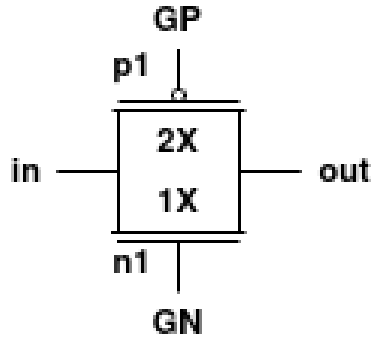


Figure 10: Transmission Gate Schematic

Device	Width (nm)	Length (nm)
pmos1v	240	45
nmos1v	120	45

Table 4: Transmission Gate Transistor Dimensions

5.2 Down-shifter

5.2.1 1.8V Inverter

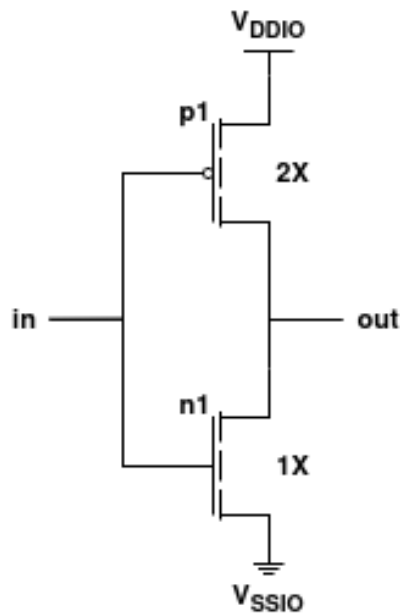


Figure 11: 1.8V Inverter Schematic

Device	Type	Width (nm)	Length (nm)
p1	pmos2v	640	150
n1	pmos2v	320	150

Table 5: 1.8V Inverter Transistor Dimensions

5.2.2 H2L Inverter

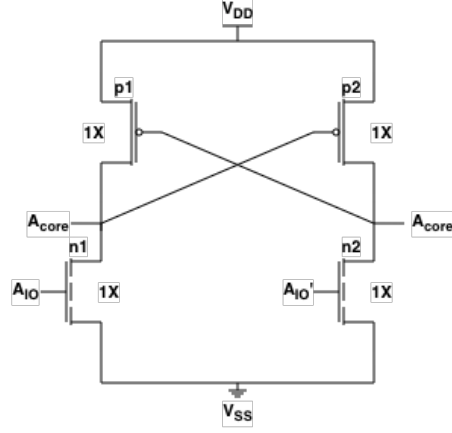


Figure 12: H2L Inverter Schematic

Device	Type	Width (nm)	Length (nm)
p1	pmos1v	120	45
p2	pmos1v	120	45
n1	nmos2v	320	150
n2	nmos2v	320	150

Table 6: H2L Inverter Transistor Dimensions

5.3 Up-shifter

5.3.1 L2H Inverter

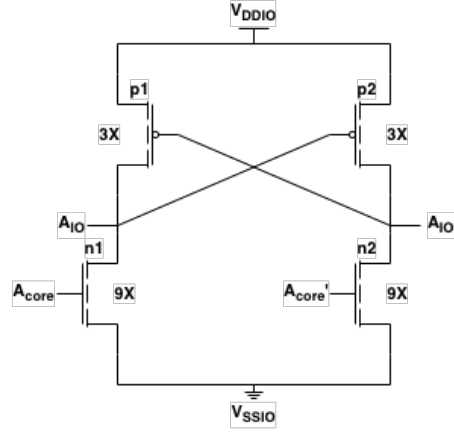


Figure 13: L2H Inverter Schematic

Device	Type	Width (nm)	Length (nm)
p1	pmos2v	960	150
p2	pmos2v	960	150
n1	nmos2v	2880	150
n2	nmos2v	2880	150

Table 7: L2H Inverter Transistor Dimensions

5.3.2 Superbuffer Inverter

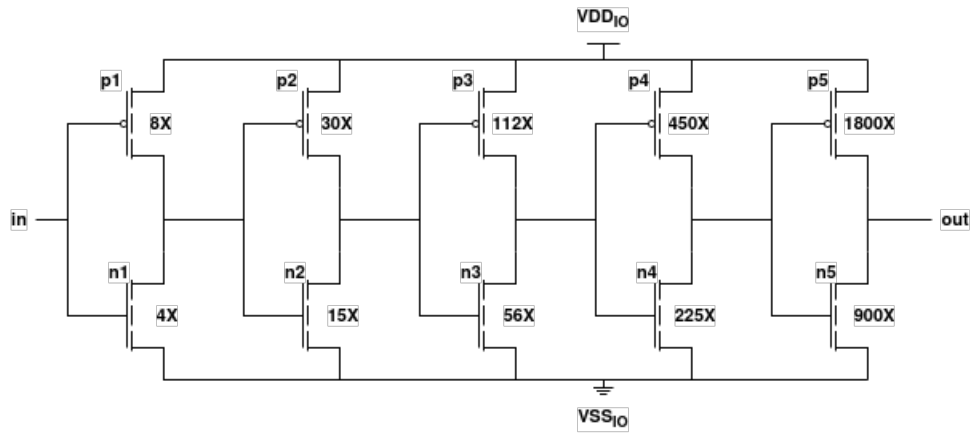


Figure 14: Superbuffer Inverter Schematic

Device	Type	Width (μm)	Length (μm)
p1	pmos2v	2.56	0.15
p2	pmos2v	9.6	0.15
p3	pmos2v	35.84	0.15
p4	pmos2v	144	0.15
p5	pmos2v	576	0.15
n1	nmos2v	1.28	0.15
n2	nmos2v	4.8	0.15
n3	nmos2v	17.92	0.15
n4	nmos2v	72	0.15
n5	nmos2v	288	0.15

Table 8: Superbuffer Inverter Transistor Dimensions

6 Results

6.1 Transient Simulations

6.1.1 Transient Waveforms at 1.1V

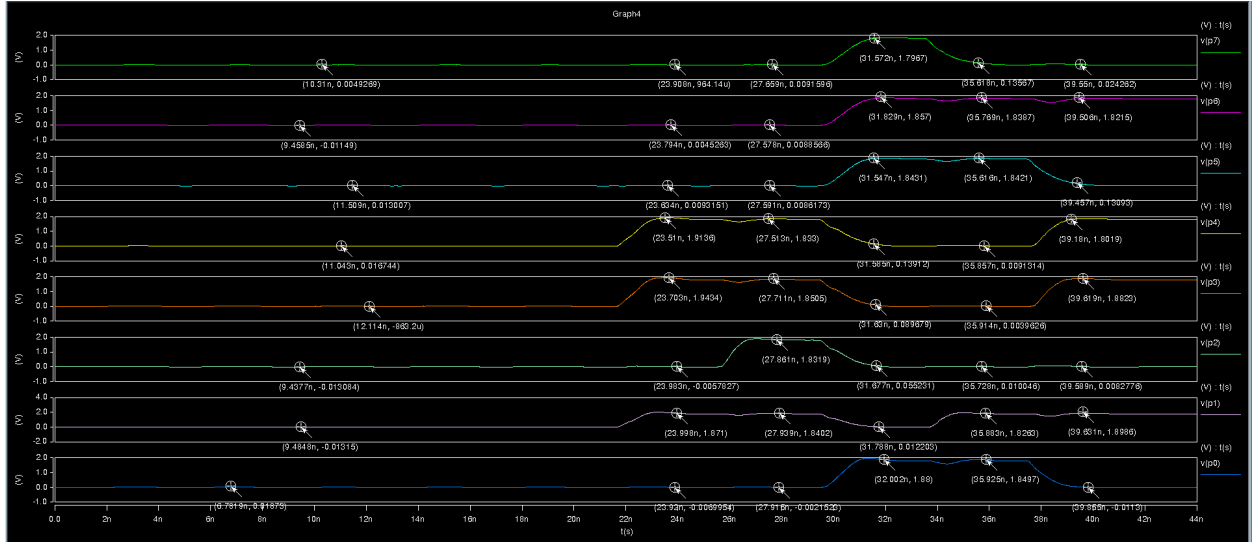


Figure 15: Transient waveforms at TT corner for Testbench 1 at 1.1V.

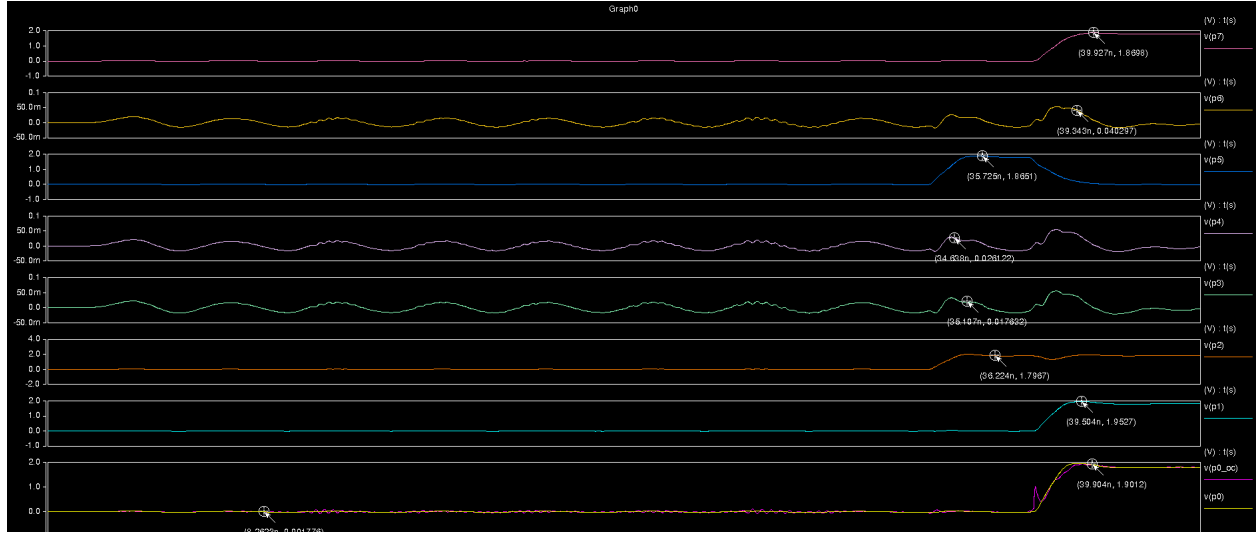


Figure 16: Transient waveforms at TT corner for Testbench 2 at 1.1V.

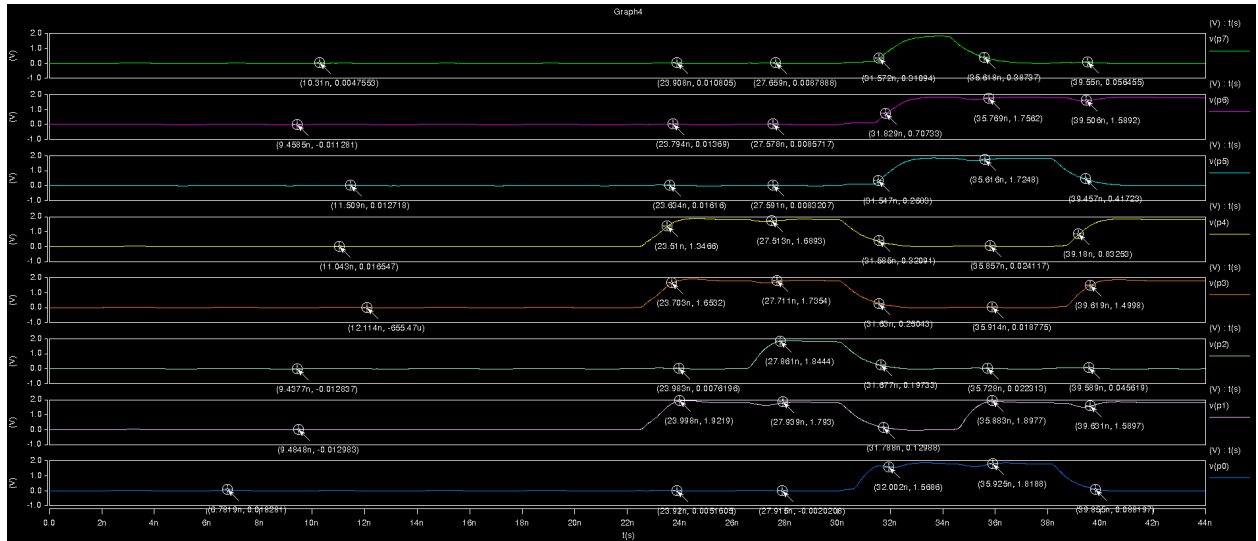


Figure 17: Transient waveforms at SS corner for Testbench 1 at 1.1V.

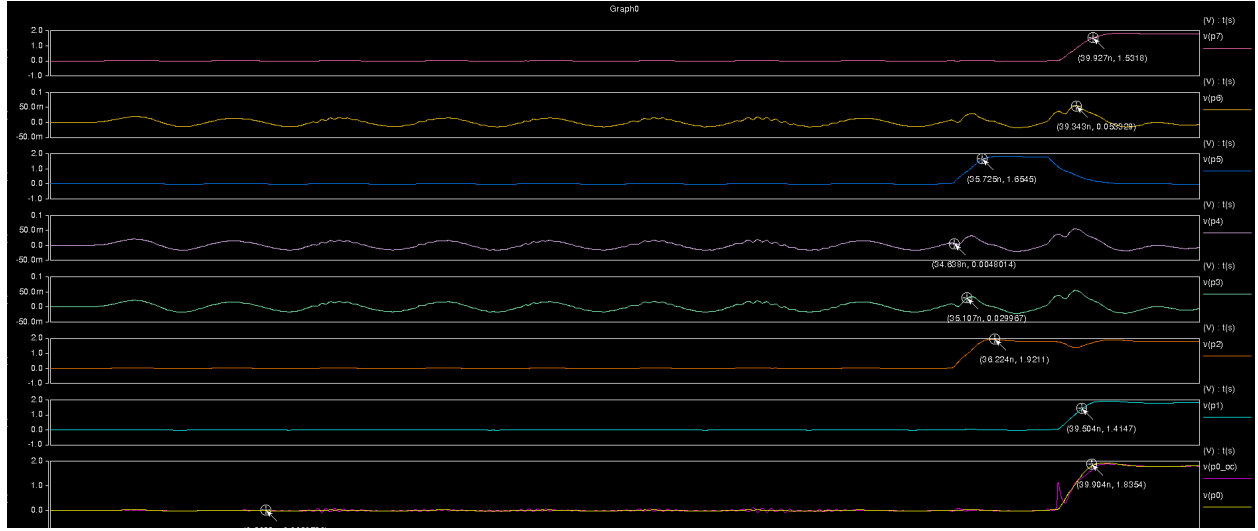


Figure 18: Transient waveforms at SS corner for Testbench 2 at 1.1V.

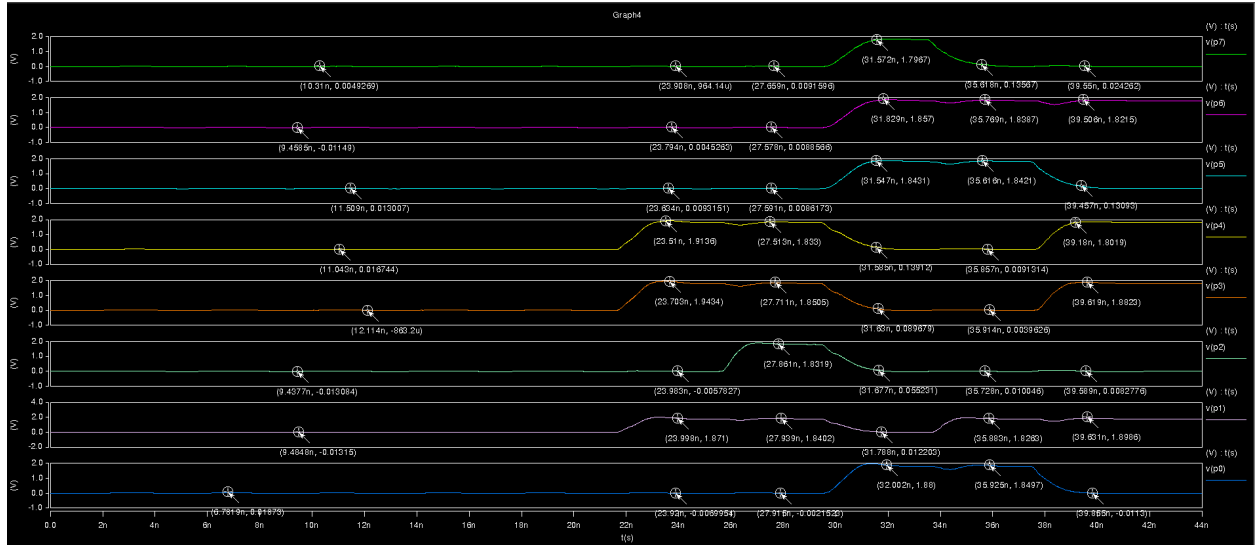


Figure 19: Transient waveforms at FF corner for Testbench 1 at 1.1V.

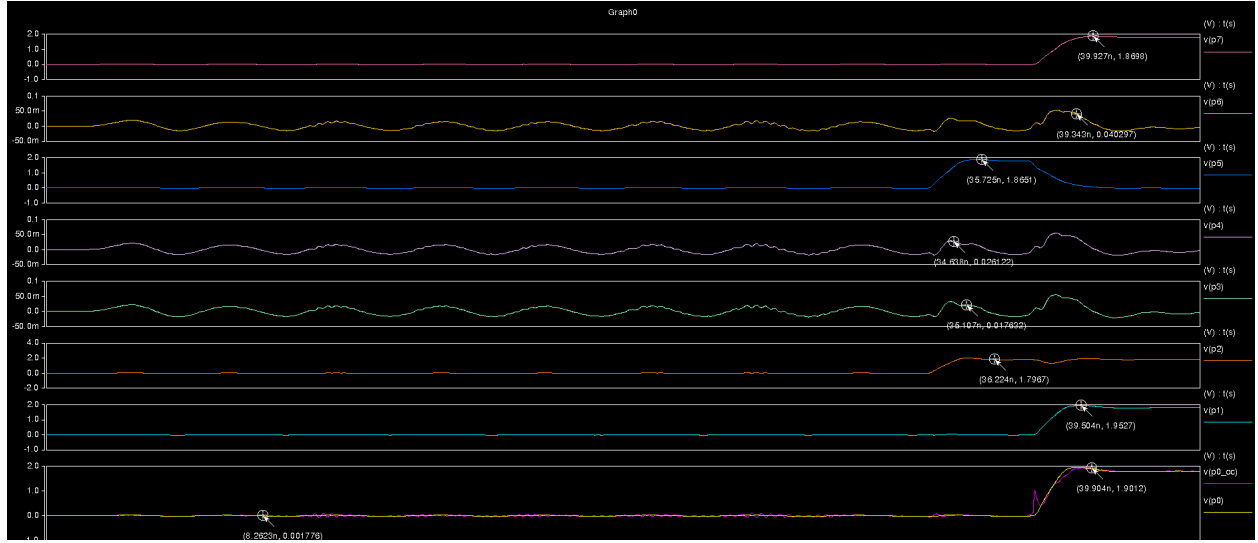


Figure 20: Transient waveforms at FF corner for Testbench 2 at 1.1V.

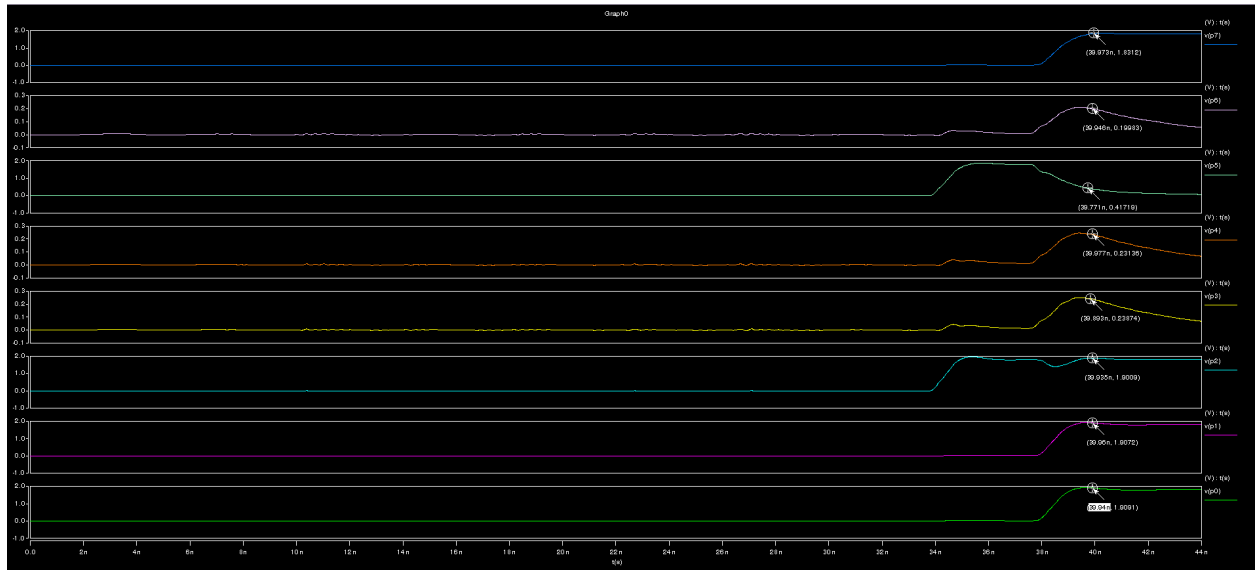


Figure 21: Transient waveforms at SF corner for Testbench 1 at 1.1V.

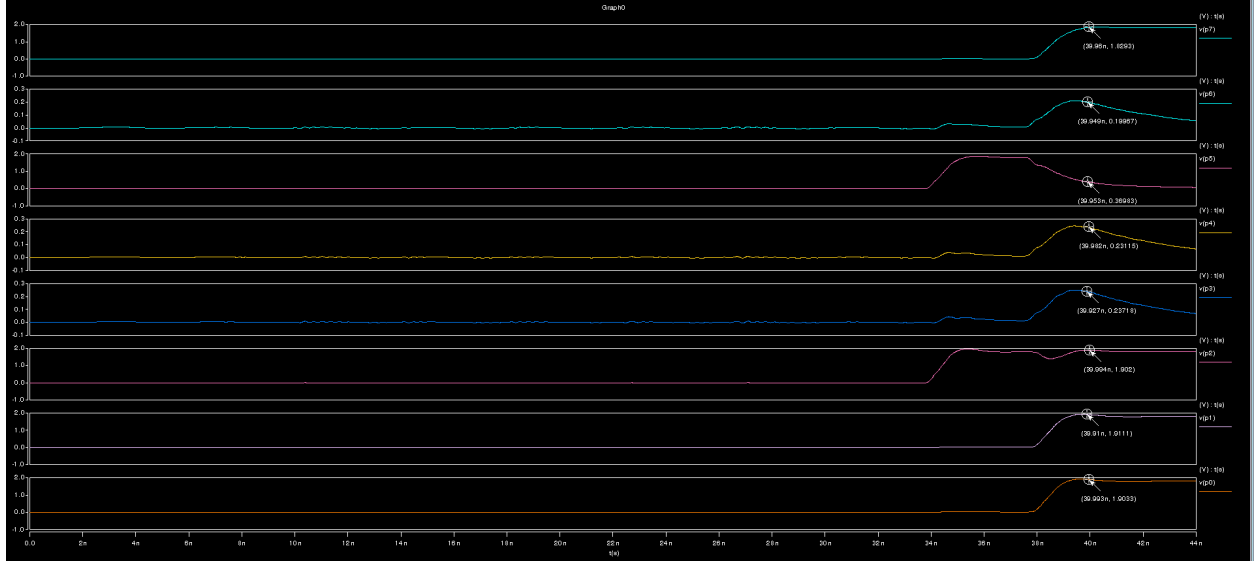


Figure 22: Transient waveforms at SF corner for Testbench 2 at 1.1V.

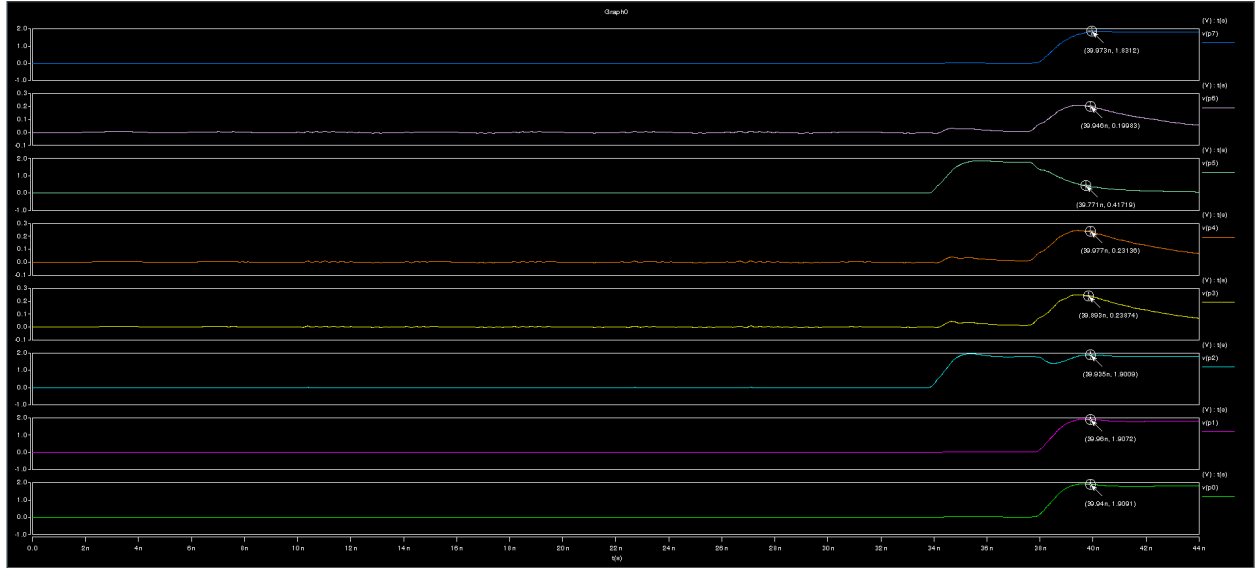


Figure 23: Transient waveforms at FS corner for Testbench 1 at 1.1V.

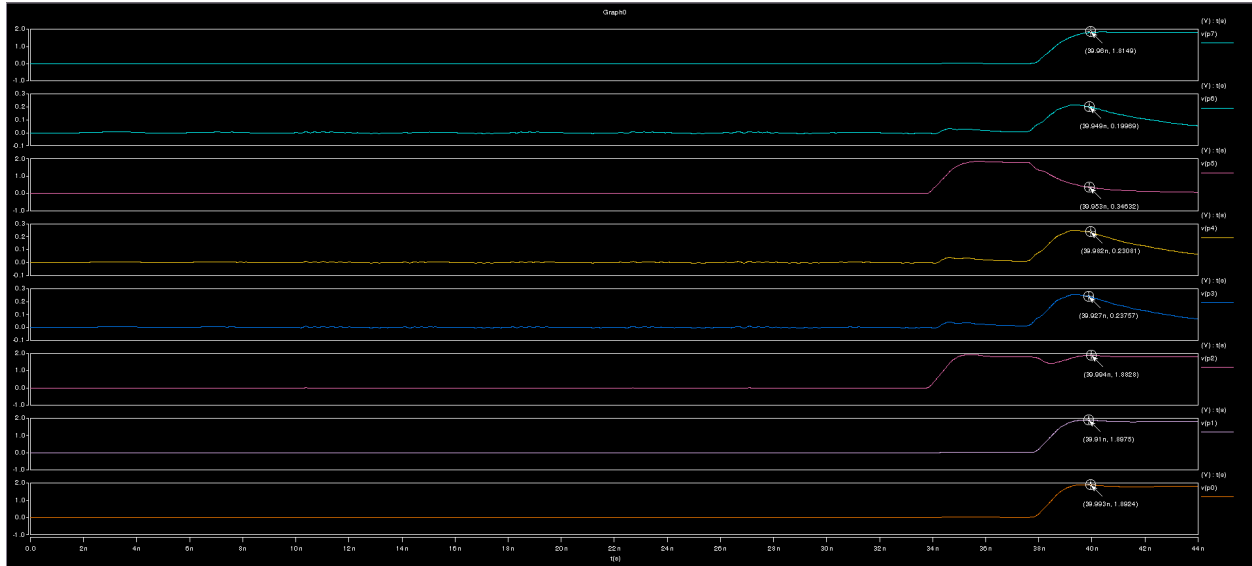


Figure 24: Transient waveforms at FS corner for Testbench 2 at 1.1V.

6.1.2 Transient Waveforms at 1.05V

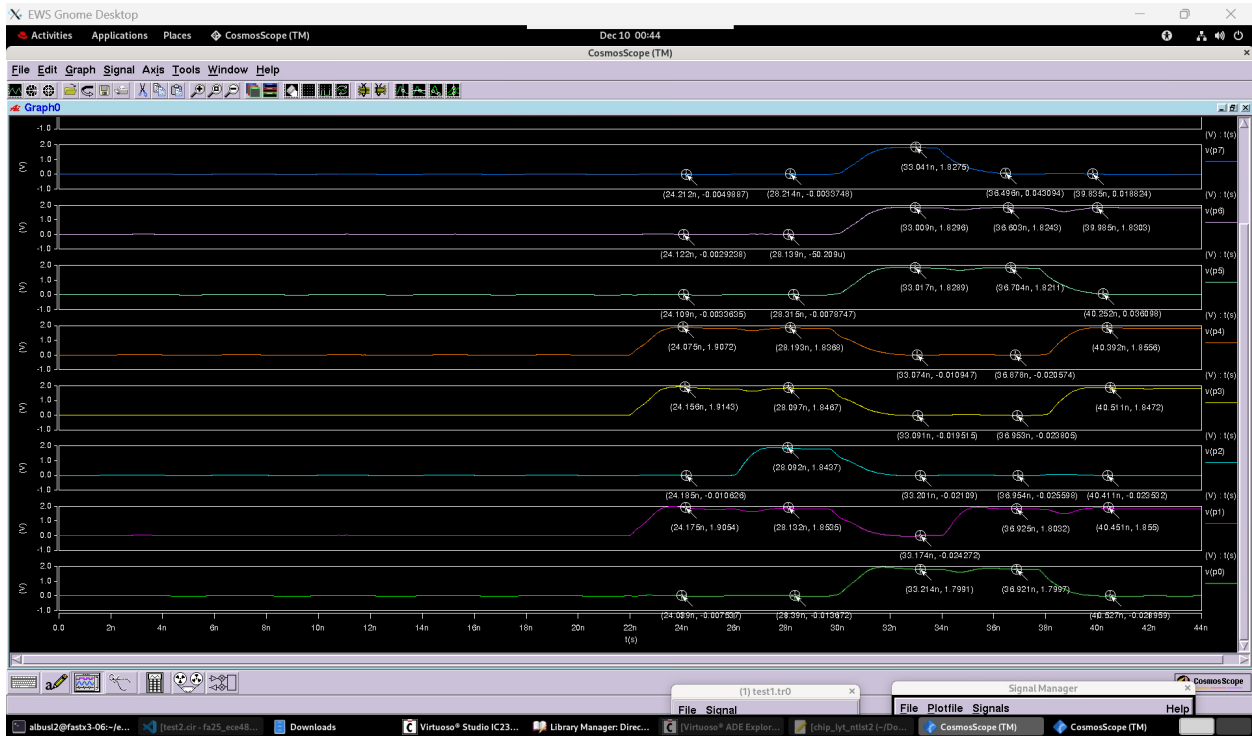


Figure 25: Transient waveforms at TT corner for Testbench 1 at 1.05V.

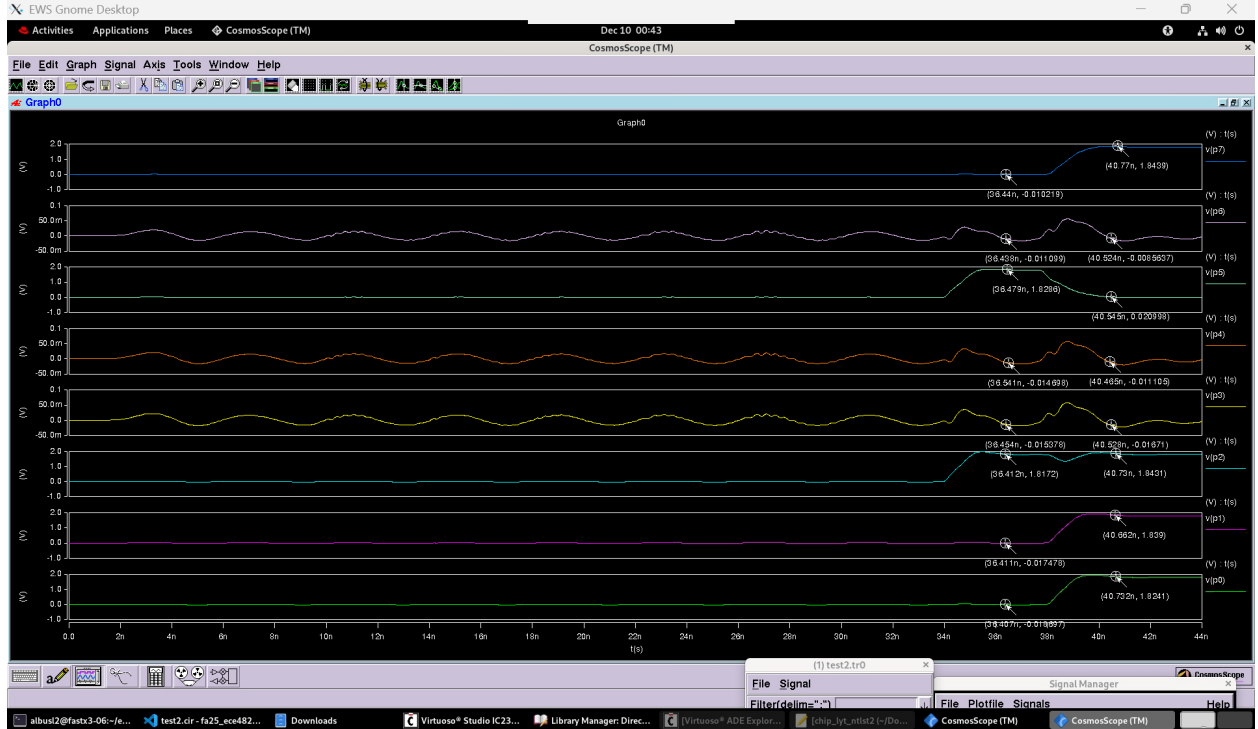


Figure 26: Transient waveforms at TT corner for Testbench 2 at 1.05V.

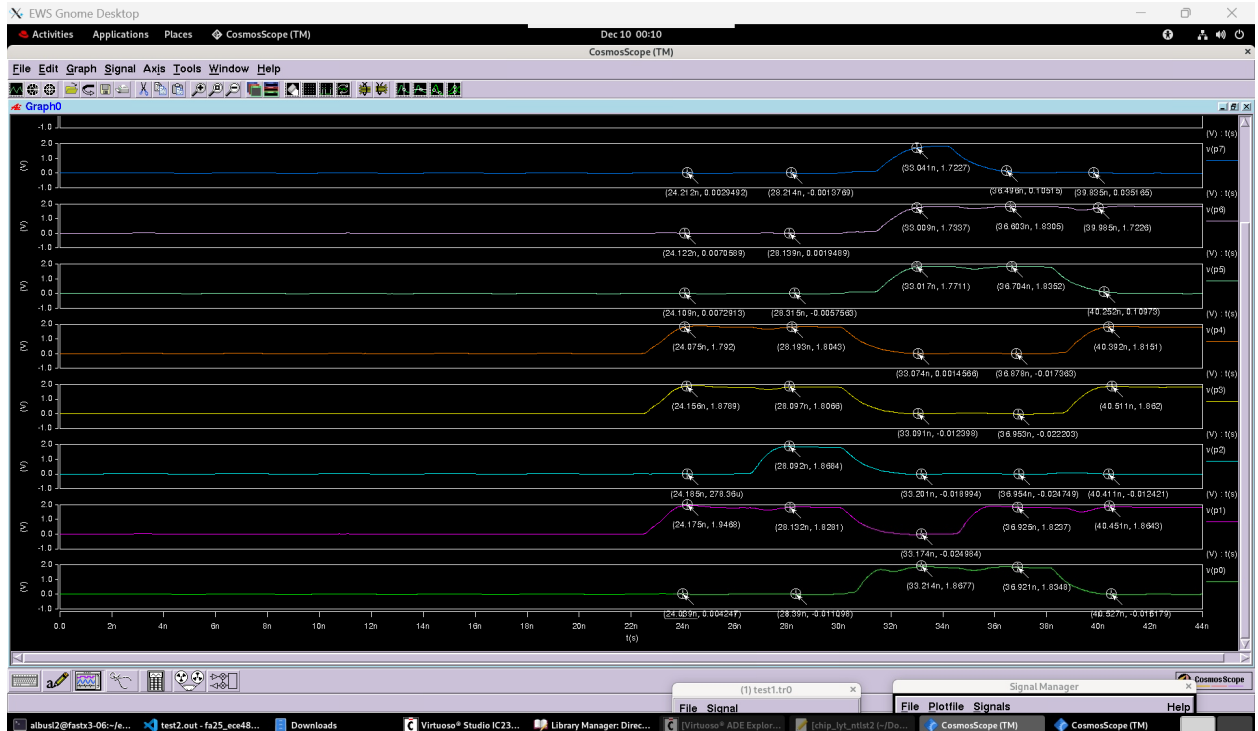


Figure 27: Transient waveforms at SS corner for Testbench 1 at 1.05V.

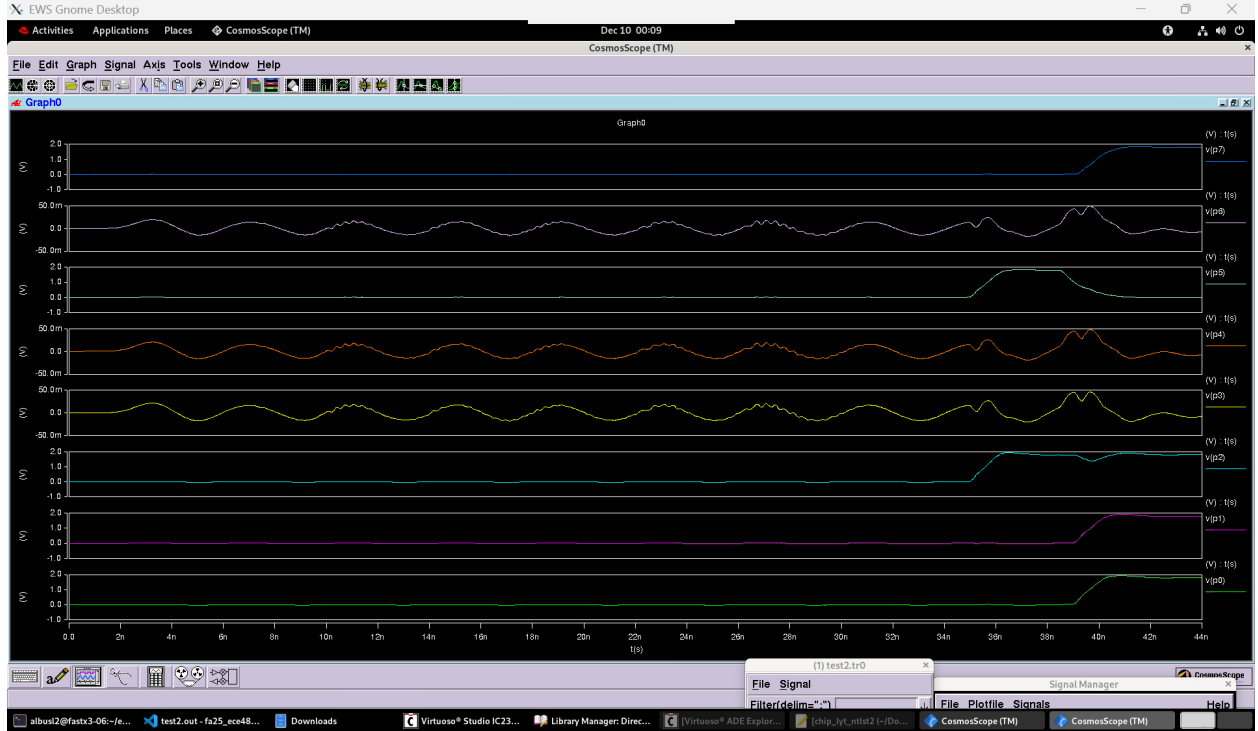


Figure 28: Transient waveforms at SS corner for Testbench 2 at 1.05V.

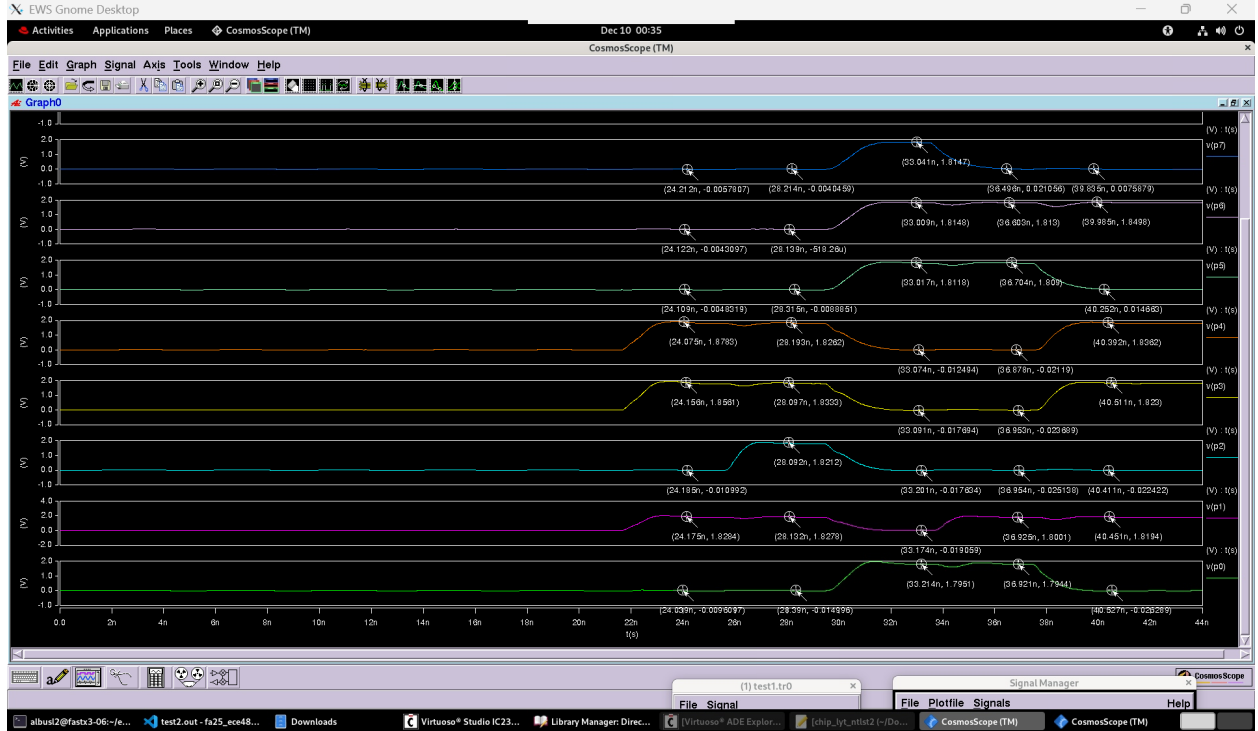


Figure 29: Transient waveforms at FF corner for Testbench 1 at 1.05V.

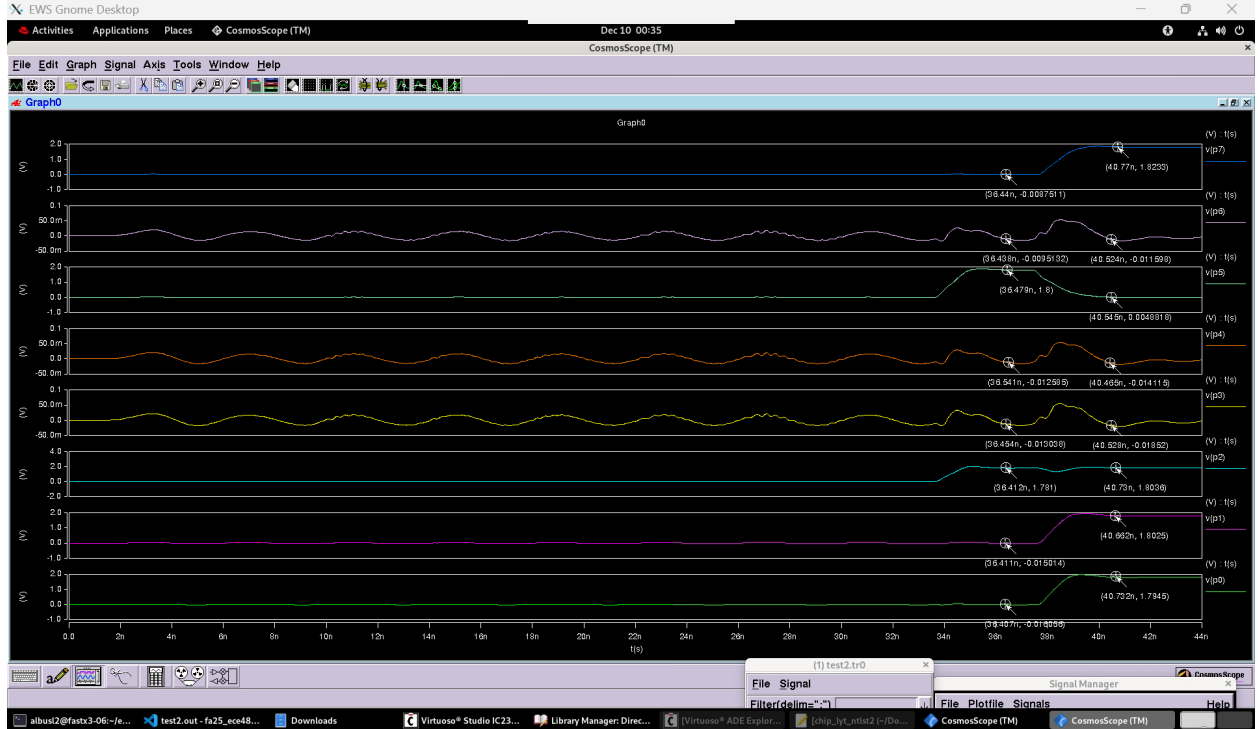


Figure 30: Transient waveforms at FF corner for Testbench 2 at 1.05V.

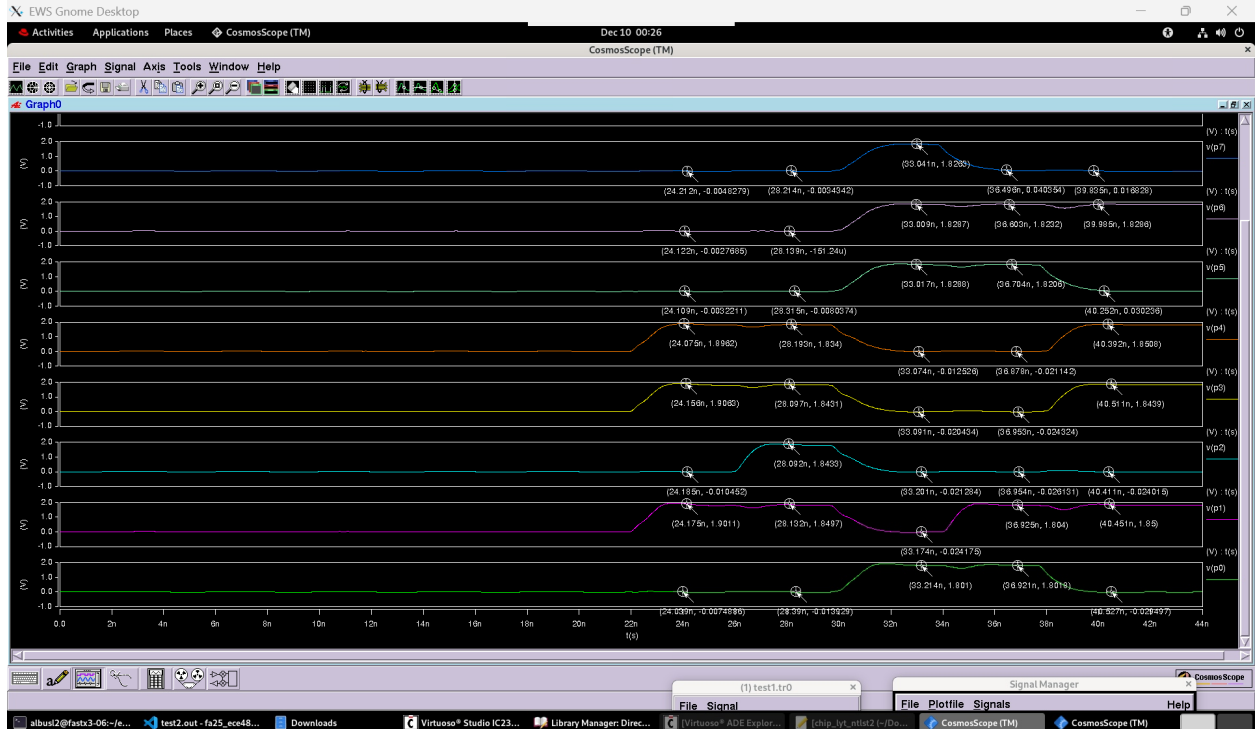


Figure 31: Transient waveforms at FS corner for Testbench 1 at 1.05V.

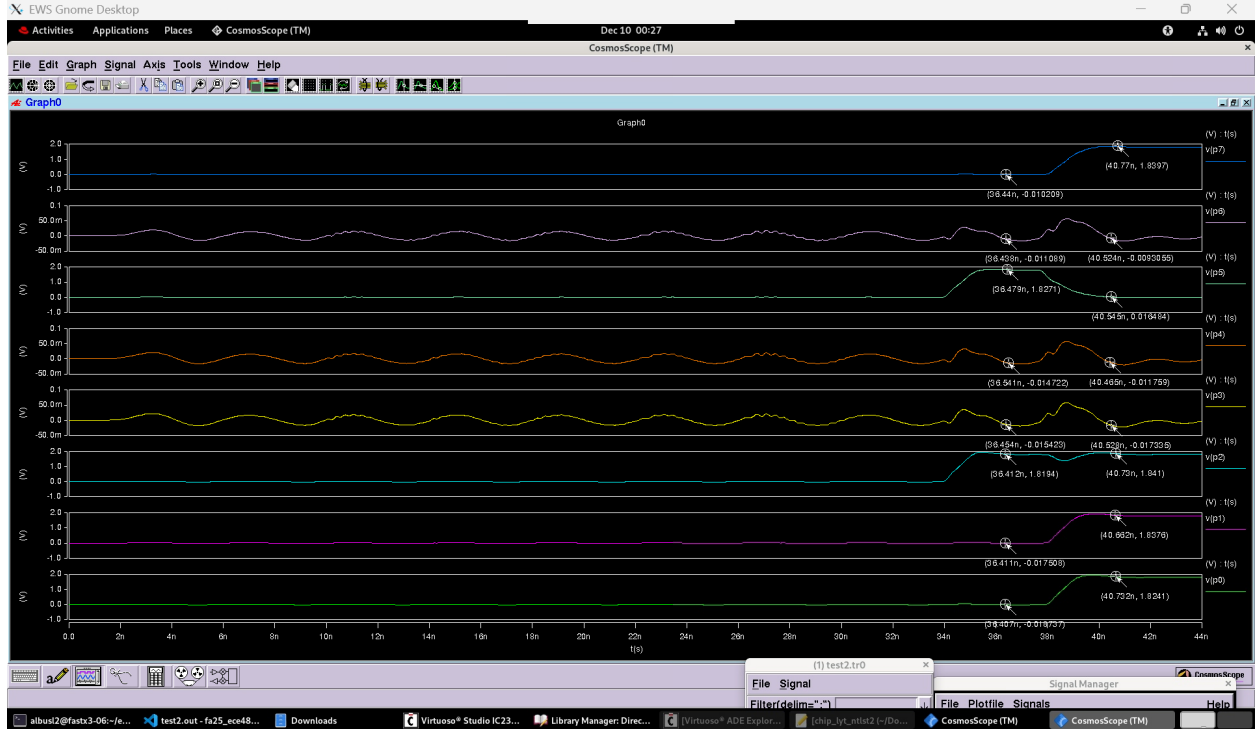


Figure 32: Transient waveforms at FS corner for Testbench 2 at 1.05V.

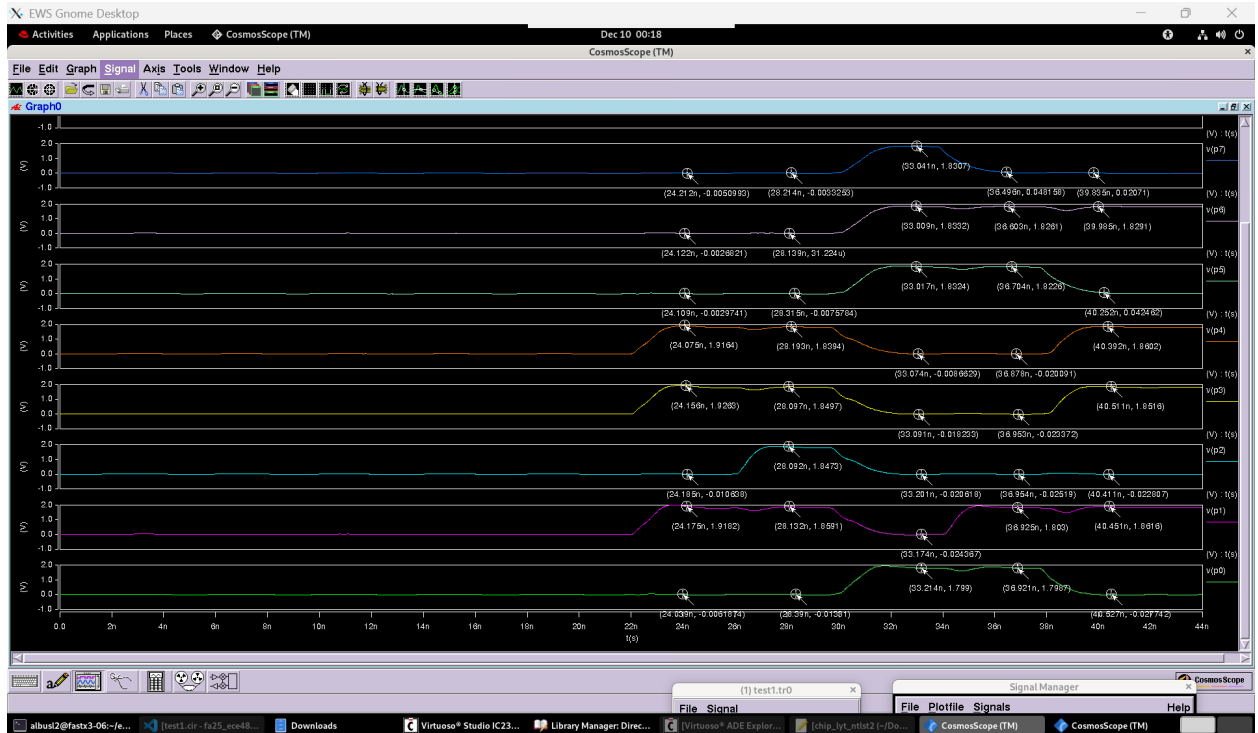


Figure 33: Transient waveforms at SF corner for Testbench 1 at 1.05V.

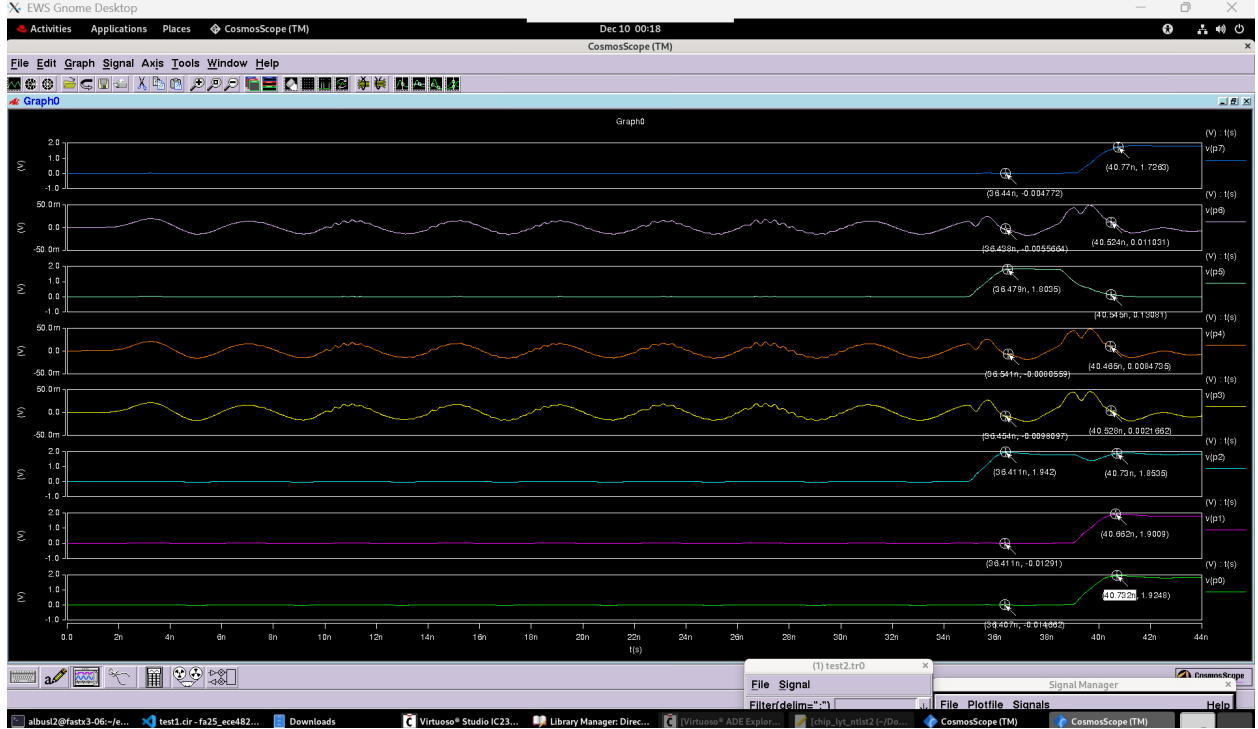


Figure 34: Transient waveforms at SF corner for Testbench 2 at 1.05V.

6.2 Power Results for 1.1V

6.2.1 Power Measurements (TT) For Testbench 1

- Average Power VDD (TT): 21.83 mW
- Average Power VDDIO (TT): 14.57 mW
- Corrected Power VDD (TT): 1.14 mW

6.2.2 Power Measurements (TT) For Testbench 2

- Average Power VDD (TT): 21.92 mW
- Average Power VDDIO (TT): 10.12 mW
- Corrected Power VDD (TT): 1.057 mW

6.3 Power Results for 1.05V

6.3.1 Power Measurements (TT) For Testbench 1

- Average Power VDD (TT) : 20.78 mW
- Average Power VDDIO (TT): 14.20 mW
- Corrected Power VDD (TT): 1.047 mW

6.3.2 Power Measurements (TT) For Testbench 2

- Average Power VDD (TT): 20.87 mW
- Average Power VDDIO (TT): 10.04 mW
- Corrected Power VDD (TT): 0.963 mW

7 Layout

7.1 Top-Levels Layout

7.1.1 Top Level

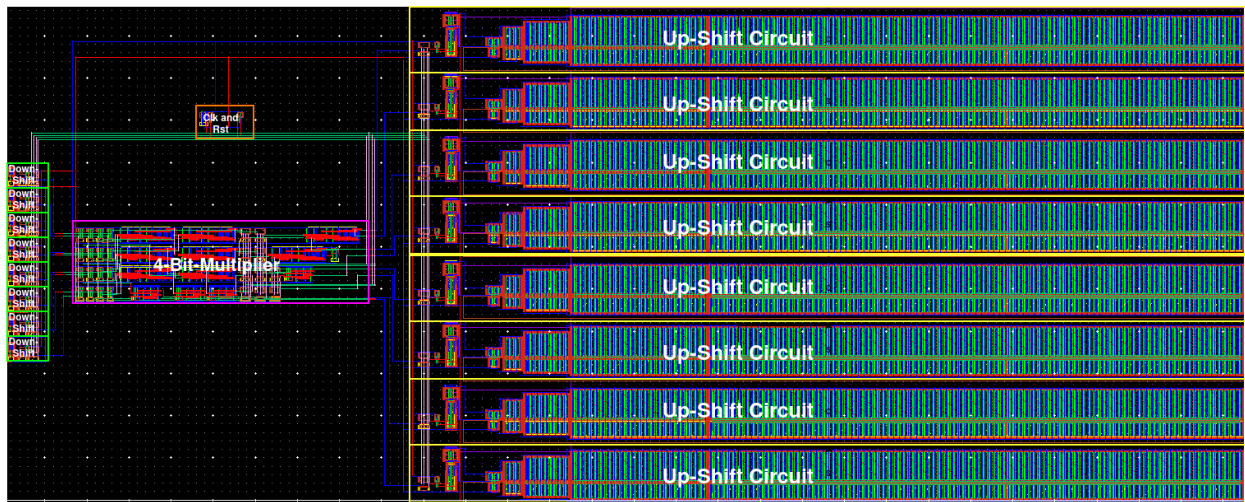


Figure 35: Top-level Layout

7.1.2 On Chip

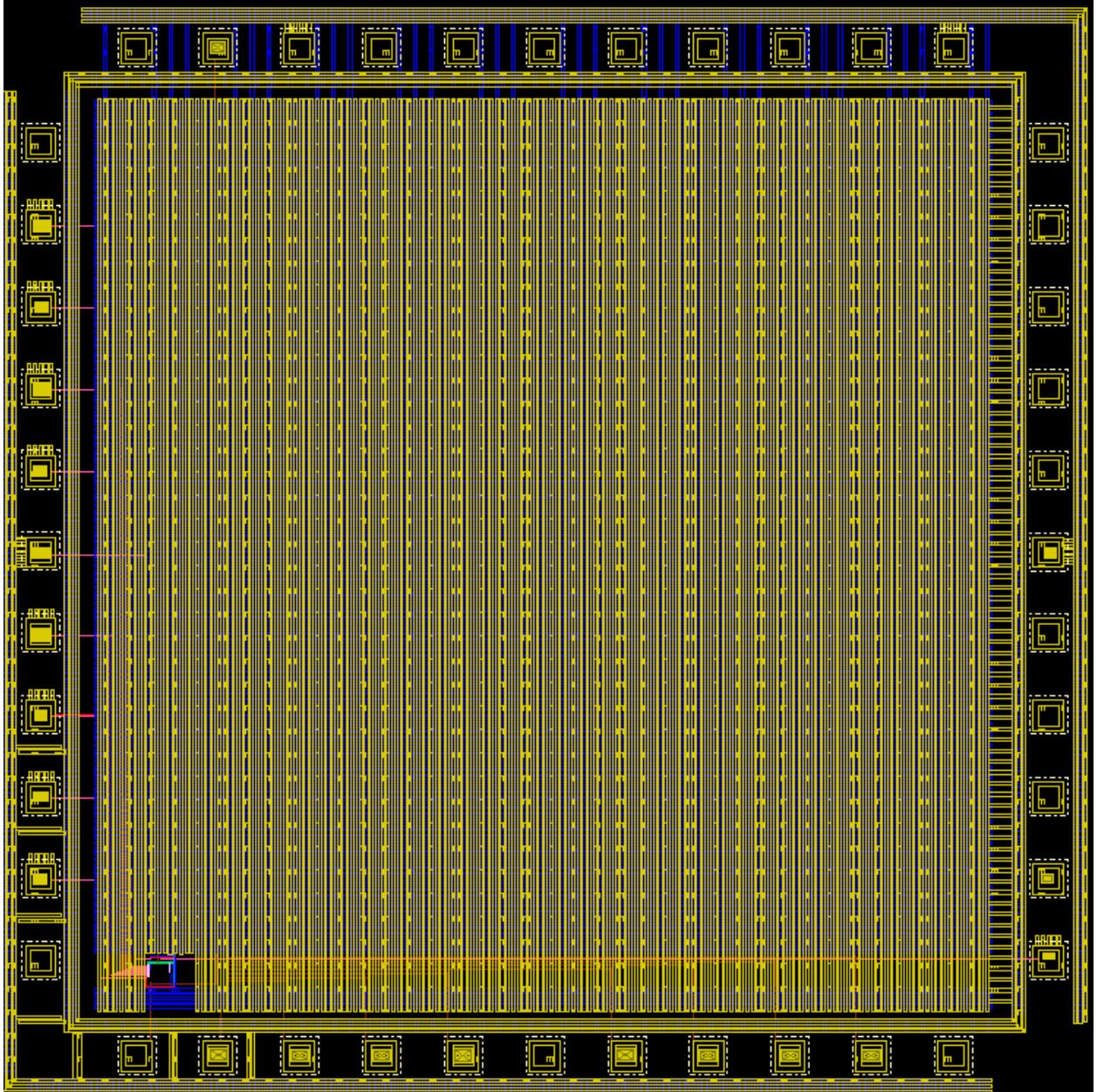


Figure 36: On Chip Layout

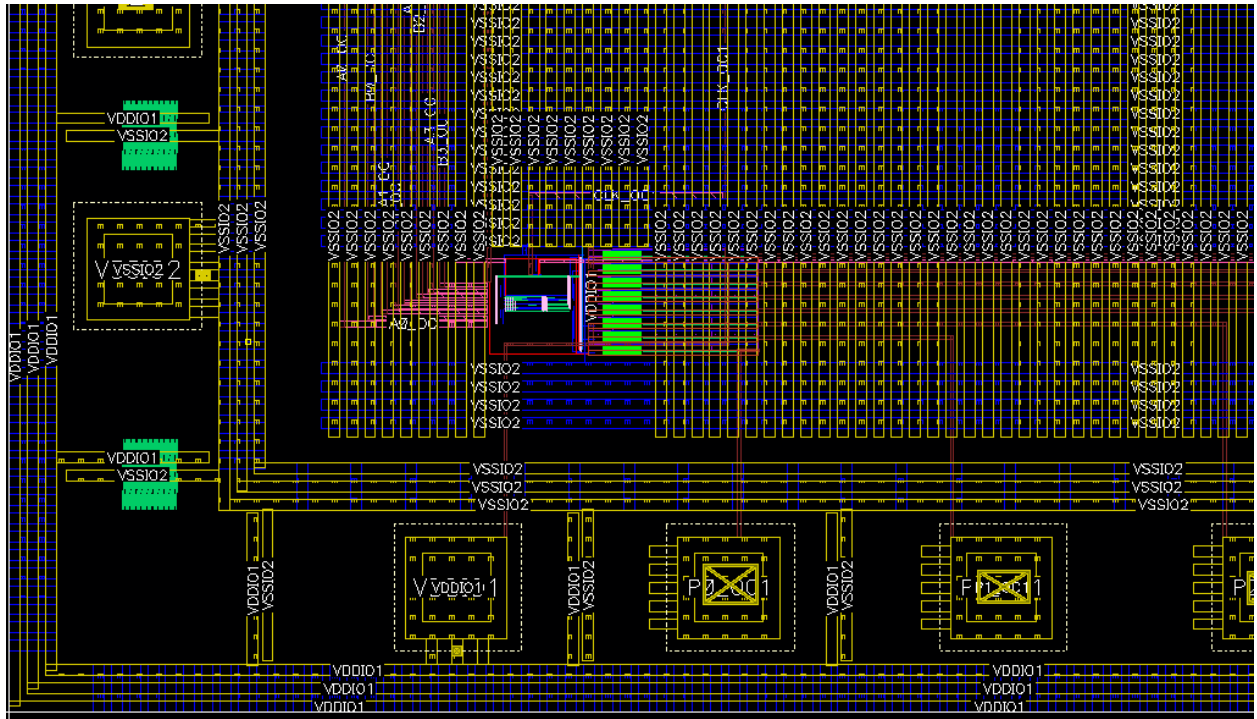


Figure 37: Closer View of On Chip Layout

7.2 Multiplier Layout

7.2.1 Half-Adder Layout

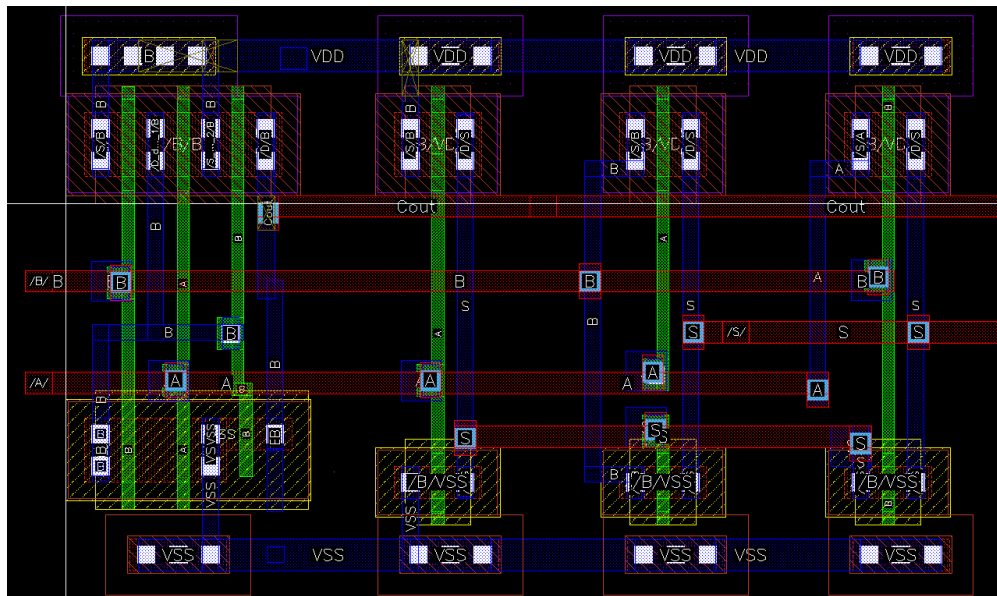


Figure 38: Half-Adder Layout

7.2.2 Full-Adder Layout

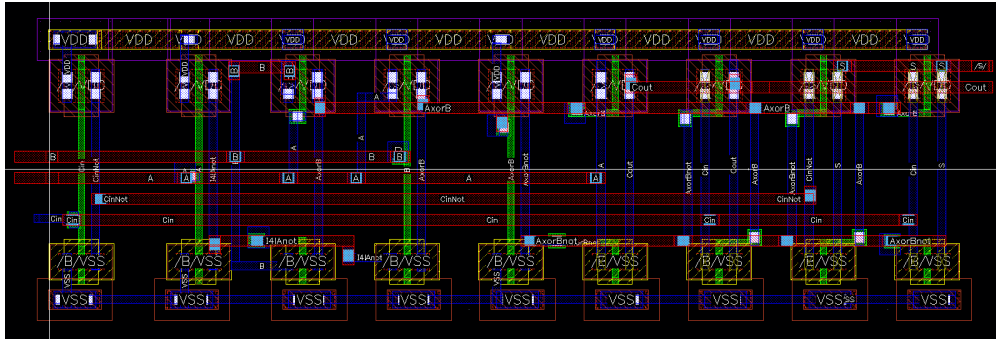


Figure 39: Full-Adder Layout

7.2.3 AND2 Layout

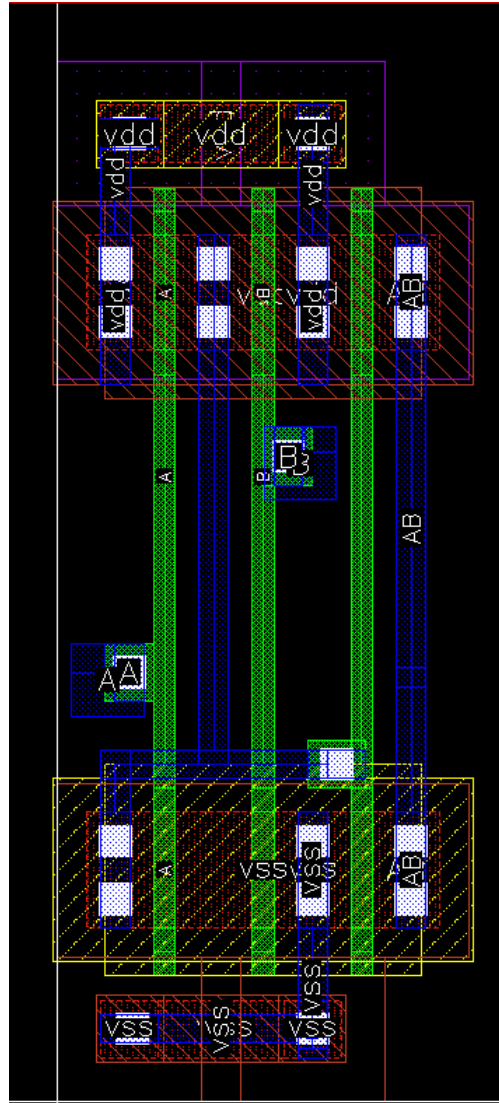


Figure 40: AND2 Layout

7.3 Down-Shifter Layout

7.3.1 1.8V Inverter Layout

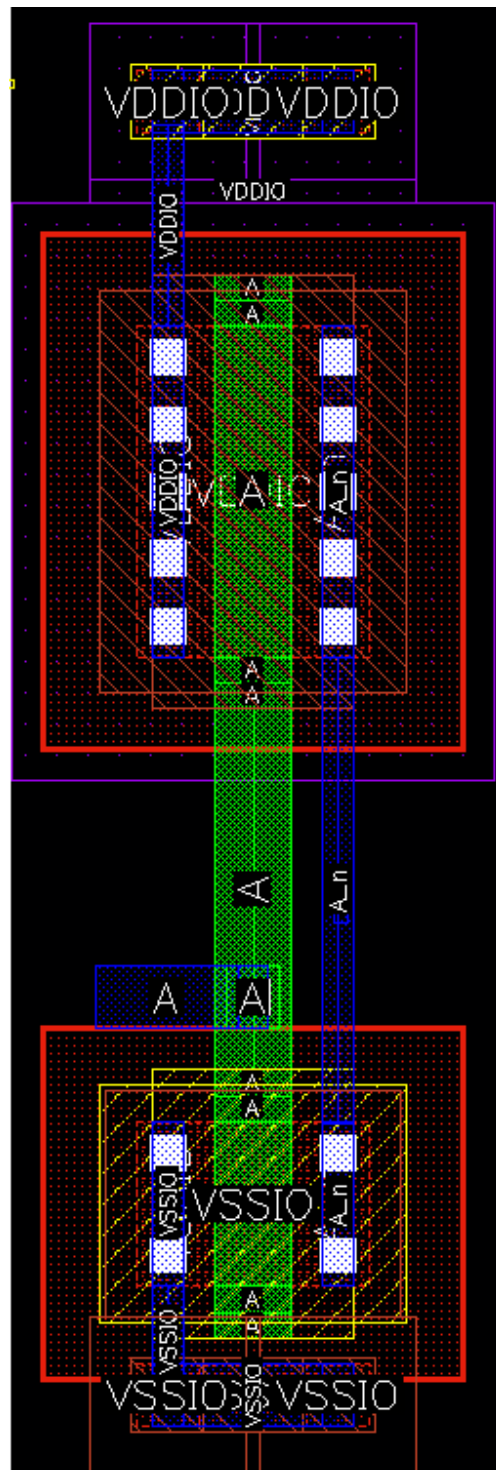


Figure 41: 1.8V Inverter Layout

7.3.2 H2L Inverter Layout

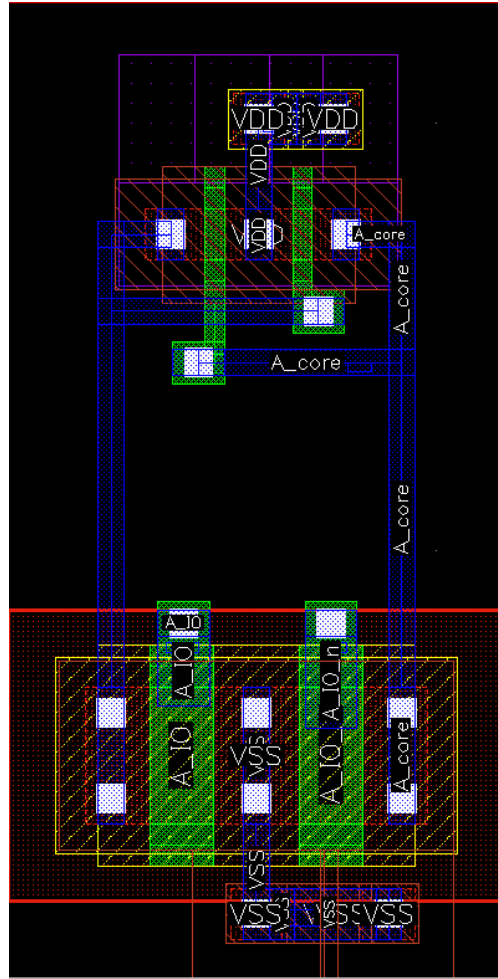


Figure 42: H2L Inverter Layout

7.4 Up-Shifter Layout

7.4.1 L2H Inverter Layout

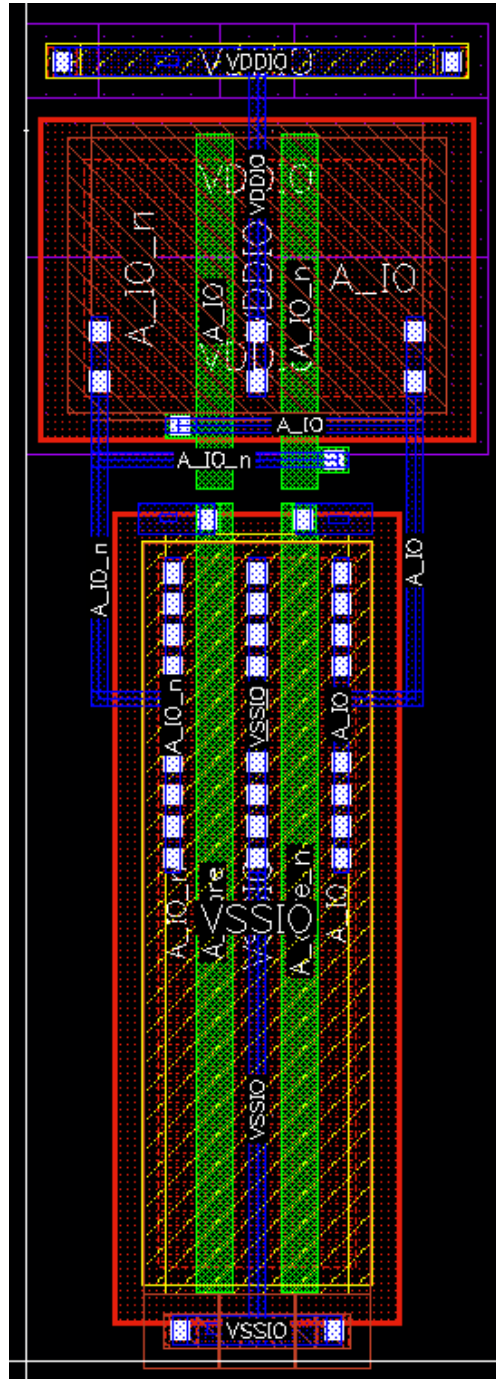


Figure 43: L2H Inverter Layout

7.4.2 Superbuffer Layout

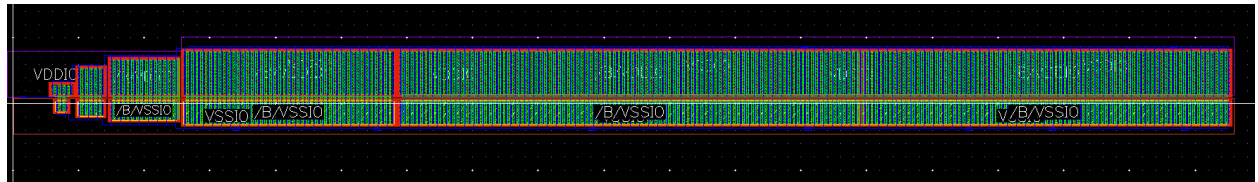


Figure 44: Superbuffer Layout

7.5 General Purpose Layout

7.5.1 Register Layout

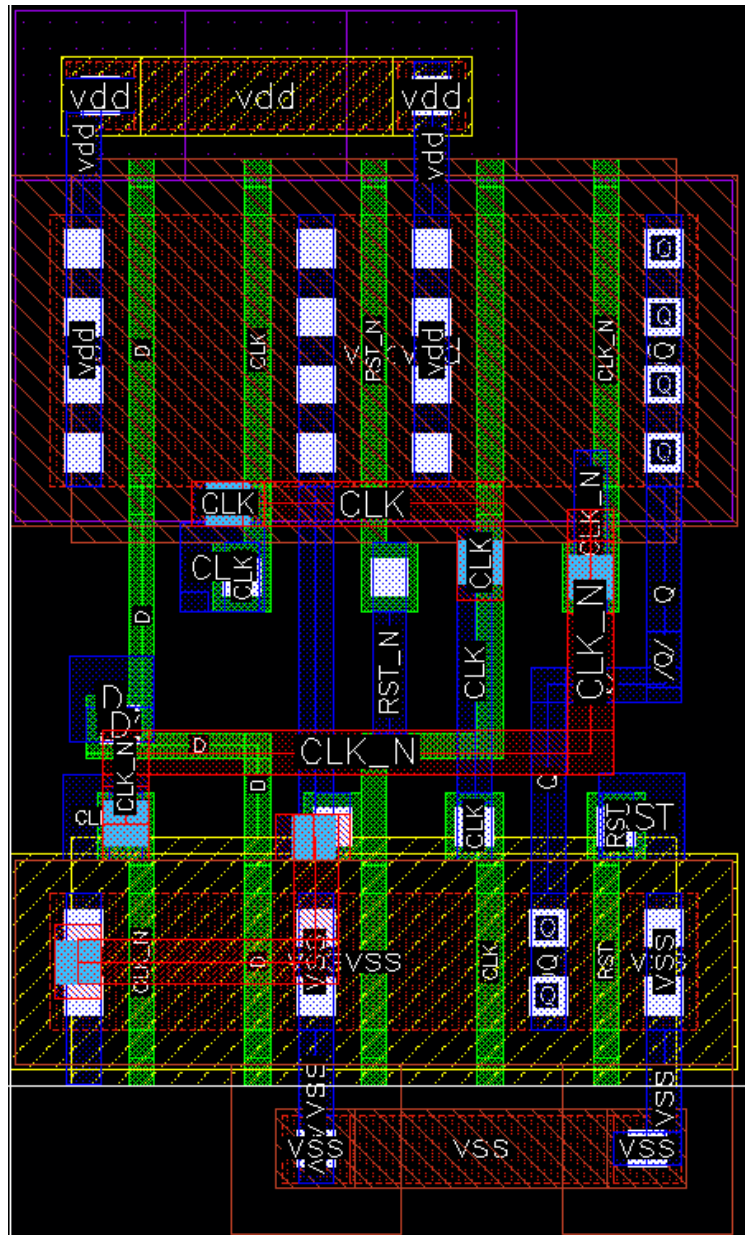


Figure 45: Register Layout

7.5.2 1.1V Inverter Layout

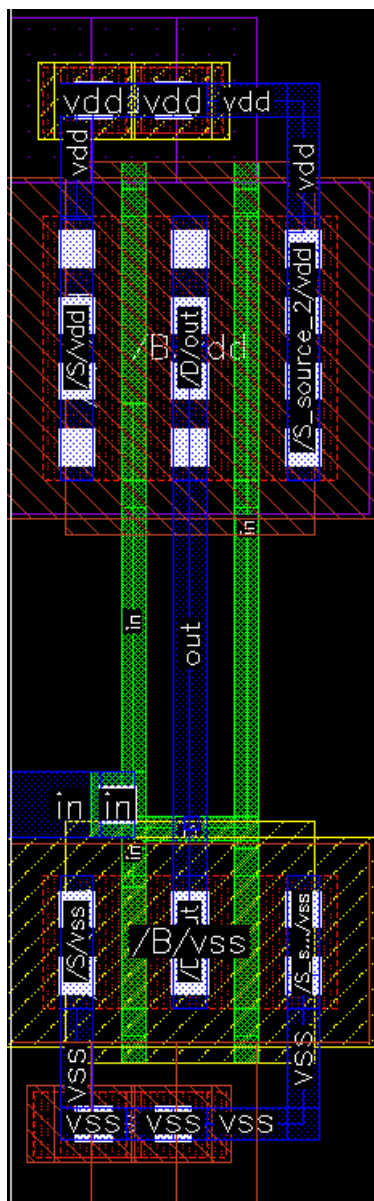


Figure 46: 1.1V Inverter Layout

7.5.3 Buffer Layout

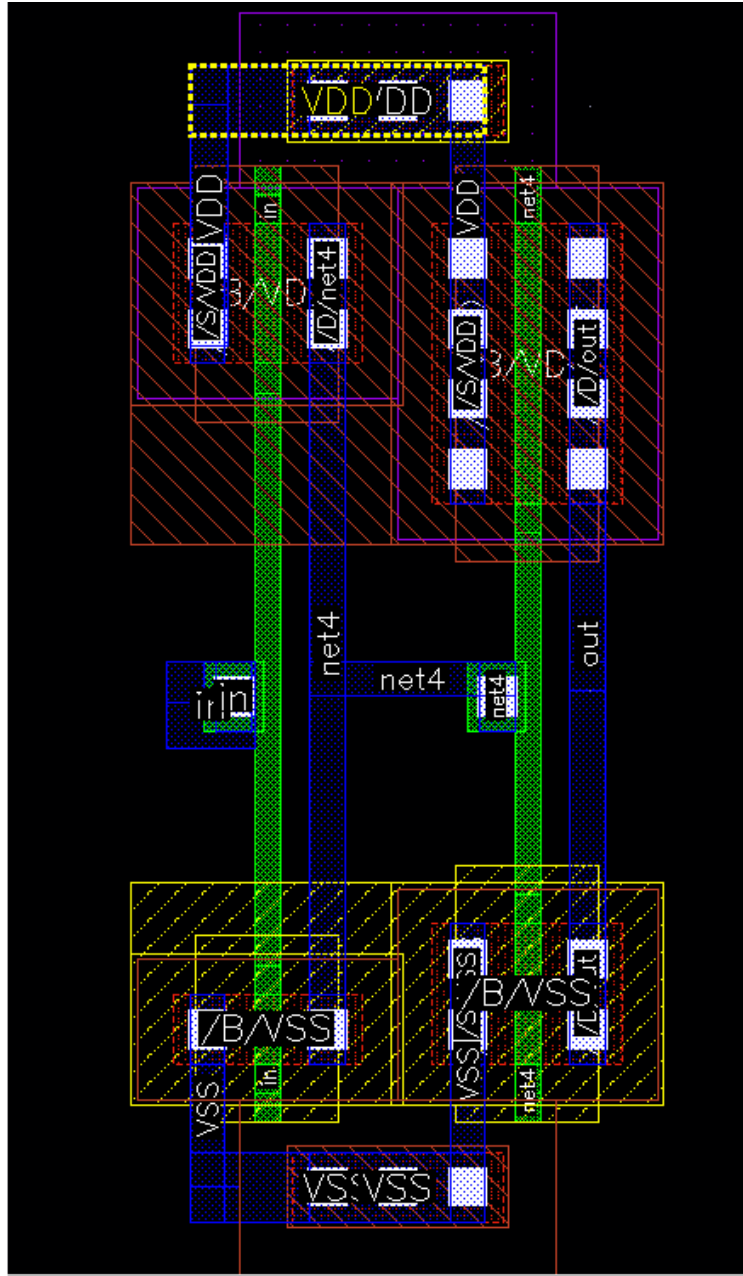


Figure 47: Buffer Layout

7.6 Total Area

- Total layout area: 8664.72 μm^2

8 Conclusions

In the end, our 4-bit multiplier design was validated to operate at 250Mhz, consuming 1mW of power, with valid outputs 3 clock cycles after the inputs are registered at 1.1V, or with valid outputs 4 clock cycles later after inputs are registered at 1.05V. For this project, we got to explore and employ many power saving strategies to minimize our designs power consumption. Strategies ranging from the micro-architecture level in picking which circuit to implement our logic, to layout strategies to minimize area and parasitic capacitance.

As for shortcomings, since we were able to meet our requirements, nothing went particularly wrong, but if we were to do things again, there are definitely different avenues to explore. More testing could of been done on the sizes in our circuits. For example, we could reduce the register size to reduce dynamic delay if it does not significantly affect timing. Additionally, the difference between schematic and layout results is much greater than we expected. For example, we were able to reduce the voltage to around 0.8V in our schematic. However, when the actual layout was used, factors like noise and resistance prevented a significant voltage drop. Our layout was also not the most optimal, as there could of been more drain source merging as well as greater consideration when it came to connecting layout together, and layout spacing especially in the on chip level. Additionally, we should have connected the gates using metal rather than the poly layer.

9 Team Effort

- Albus L. — 4-Bit multiplier design (schematic and layout of all parts), help with top level and on chip layout, DRC and LVS fixing, testbenching, and data collection for report.
- Ashton B. — Up and Down Shifter circuit design (schematic and layout of all parts), top level and on chip layout, report writing and management.
- Chirag M. — Help with the superbuffers layout, report writing and management.